TfL What-If Simulator

User Guide and Demonstration Script

Step-by-step guide for using and presenting the implemented system

Document Structure

1. Audience and Purpose

This document explains how to use the implemented TfL What-If Simulator and provides a structured demonstration script. It focuses on the visible system behaviour and how to present it clearly.

2. Quick Start

1. Open the web application and wait for the Underground map to load.
2. Use the collapsed search control if you want to jump to a specific station.
3. Click one station to set a route start, then click another station to set the destination.
4. Use the main-screen What-If toggle to enter simulation mode.
5. Close stations or lines, then plan a route again to show how the network adapts.
6. Save the scenario if you want to reload the same disruption state later.

3. Using the Map

The map is the main workspace. Users can pan and zoom like a normal web map. Stations and Underground connections are rendered as interactive map elements. Station markers can be clicked for route planning or What-If closure interaction depending on the current mode. A HUD/control area provides reset and route-management actions.

Action	Expected behaviour
Pan map	Move around the London network while keeping the map constrained to the useful area.
Zoom map	Increase or decrease map detail. Station marker sizing adapts to zoom level.
Click station in route mode	Select start or destination station depending on current route state.
Click station in What-If mode	Toggle closure state or use contextual route actions depending on interaction mode.
Reset view	Return the map and route state to a clean baseline.

4. Planning Routes

To plan a route, select a start station and then select a destination station. The system calculates a path through the graph and highlights it on the map. The route panel summarises the route, estimated travel time and interchange information. After the route is found, the map adjusts so the selected route can be seen clearly.

The route algorithm uses travel time weighting and a penalty for line changes. This means it is not simply counting the number of stations; it is trying to produce a more realistic route by considering travel time and the cost of changing line.

5. Using What-If Mode

What-If mode is the simulation mode. It allows the user to model hypothetical disruptions without changing the live network graph. When enabled, simulated closures take priority over live closure state. The mode is intentionally visible from the main screen so users do not have to search through the sidebar to find it.

What-If action	Purpose	Result
Enable What-If toggle	Enter simulation mode.	The UI changes to a simulation context and closure controls become active.
Close station	Model a station outage or inaccessible station.	The station is marked as closed and route planning avoids it.
Close line	Model a full line suspension.	Routes ignore all edges belonging to that line.
Reset closures	Return simulation to no user-defined closures.	Closed station and line sets are cleared.
Disable What-If	Return to live/normal mode.	Simulation closures no longer affect route planning.

6. Saving and Loading Scenarios

A scenario is a named snapshot of the current What-If closures. It stores closed station IDs, closed line IDs and metadata such as creation/update time. Scenarios are saved in the browser using localStorage as JSON. This means they persist after refresh on the same device/browser, but they are not synced to other devices.

7. Turn on What-If mode.
8. Close one or more stations or lines.
9. Enter a scenario name.
10. Save the scenario.
11. Refresh or continue using the app.
12. Select the saved scenario to load it again.

This approach is suitable for assessment because it demonstrates persistence without requiring user accounts, authentication or additional database schema. It also protects the Neo4j transport graph from being modified by temporary simulations.

7. Using Live Data Features

When What-If mode is off, the application can display live service context from TfL APIs. This includes line status/delay information, live or fallback train movement and live station closure awareness. If TfL data is unavailable, the application should continue to operate with fallback values rather than failing completely.

Feature	How to demonstrate it
Line status	Open the controls and show the status labels beside Underground lines.
Delay information	Hover/tap a line with status details to show reason/additional information if available.
Train display	Enable train visibility and show markers moving along the network.
Feed status	Point out whether the display is using live TfL arrivals or schedule fallback.

8. Troubleshooting

Issue	Likely explanation	What to check
Map does not show stations	/api/stations failed or Neo4j connection issue.	Check Neo4j environment variables and API response.

Routes do not calculate	Start/end not selected, graph not loaded or disruptions block route.	Clear route, reset closures and try two connected stations.
Saved scenarios disappear	Browser storage cleared or different browser/device used.	Check same browser and localStorage availability.
Train feed shows fallback	TfL arrivals unavailable or insufficient trackable vehicle IDs.	This is expected fallback behaviour.
Controls overlap	Viewport too small or layout offsets need adjustment.	Test desktop and mobile breakpoints.